What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Measuring Cardiac Function

Last updated January 28, 2026 · 13 min read · ScholarRx

Adapted by: Ian Murray, Diptiman Bose, Ryan Downey, Phil Riley, Eleanor Schrems, Patrick Brooks

Editors: Jordan Finn, Russell Johnson

Adapted from: Measuring Cardiac Function

Listen to this Brick
0:00 / 22:50 1x

Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Differentiate between stroke volume and ejection fraction and calculate both.
2. Explain methods of measuring cardiac output, including the Fick's principle, dye dilution, and thermodilution methods.

CASE CONNECTION

TU, a 55-year-old female with a history of type 2 diabetes mellitus and hypertension, presents to the primary care clinic one week after hospitalization for acute dyspnea. She recalls little from her admission but reports starting new medications. She now feels well, with no dyspnea or edema.

Hospital records show an estimated cardiac output (CO) of 3.5 L/min and an ejection fraction (EF) of 45%, leading to a diagnosis of heart failure with reduced ejection fraction (HFrEF). She was started on a beta-blocker and an ACE inhibitor. On examination, her vitals are stable, and her heart and lung exams are unremarkable. She asks for clarification on her medication and diagnosis, particularly regarding CO and EF.

How would you respond to her questions? Consider this as you read, and we'll revisit TU at the end of the brick.

GO TO CONCLUSION ↓

Introduction

Cardiac output (CO) is the primary measure of the heart's pumping efficiency. It represents the amount of blood the heart delivers throughout the body to supply oxygen and remove waste products. At times, it is important to measure cardiac output, along with related factors of stroke volume, heart rate, and ejection fraction, to determine if the heart can adjust blood flow to adequately meet the body's changing needs. Let's take some time to briefly review EDV, and SV before we move onto ejection fraction (EF).

A young patient has an HR of 70 bpm and an SV of 70 mL. What is her CO in mL/min or L/min?

How Do We Calculate Stroke Volume and Ejection Fraction (LO1)?

There are two factors involved in cardiac output: heart rate and stroke volume.

Heart Rate (HR): HR is readily determined by taking the pulse, or using the QRS frequency (R-R interval) from an electrocardiogram (ECG) when the pulse is faint (as in patients with hypotension).

Stroke Volume (SV): What about SV? The SV can be calculated as the difference between the end-diastolic volume (**EDV**) and the end-systolic volume (**ESV**).

SV = EDV – ESV

How do changes in EDV and ESV alter Stroke volume (SV)?

End-diastolic volume (EDV) is a measure of the volume of blood in the ventricle after filling, and thus preload, and it increases with the following:

- **Heart rate:** Lower heart rate (more time to fill the ventricles)
- **Venous return:** Increased blood volume or venous pressure (increased return of blood to the heart.

End-systolic volume (ESV): is the volume of blood remaining in the heart after contraction or systole. Alterations in ESV impact the SV. Factors that may alter ESV include:

- **Contractility:** Stronger or longer ventricular contractions (shift of the End-Systolic Pressure Volume Relationship (ESPVR) upwards [for review - *see Pressure Volume (PV) Loops brick*]) leads to more blood being pumped out, a lower ESV (shifted to the left), and greater SV (Figure 1).
- **Afterload:** Is increased with outflow obstruction (e.g., aortic stenosis), or arterioconstriction or stiffness (high blood pressure). Increased afterload leads to less blood leaving the heart, a higher end-systolic volume (ESV), and a smaller stroke volume (SV).

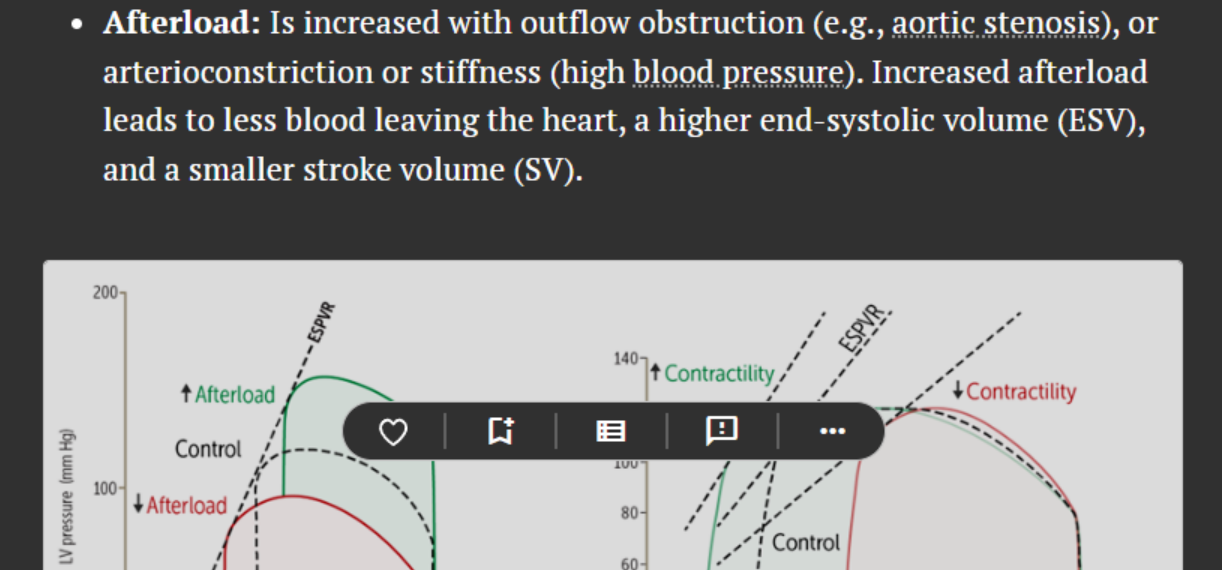

Figure 1. Visualization of how end-systolic volume (ESV) is affected by afterload and contractility

CLINICAL CORRELATION

With **aortic stenosis** or **hypertension** and increased afterload, there is increased outflow obstruction or higher resistance against which the heart must pump. This leads to more blood being retained the heart, and a higher end-systolic volume (ESV) or less blood pumped out of the heart. The reduced stroke volume (SV), and decreased cardiac output (CO), with the patient potentially having symptoms of fatigue, exercise intolerance, and in severe cases, signs of heart failure.

If EDV is he ESV?

Ejection Fraction (EF)

The ejection fraction (EF) is derived from SV and serves as a key clinical indicator of systolic function, particularly in heart failure. While stroke volume (SV) varies based on body size, EF provides a normalized measurement. EF is the percentage of blood ejected during contraction relative to the individual's specific end-diastolic volume (EDV). This is calculated as follows, and note that SV = EDV - ESV:

$$EF = SV/EDV \times 100$$

The EF is normally about 55 - 60%. It is usually measured by echocardiography.

CLINICAL CORRELATION

Heart failure with reduced EF (HFrEF) occurs when the heart does not pump as strongly (diminished less than normal (< 55%).

Heart failure with preserved EF (HFpEF) occurs when the heart is stiff and cannot fill properly (low EDV). Here, the EF is preserved, even increased, but the SV is low because there is less filling (EDV) at the start of contraction (both SV and EDV go down). The cardiac output is low under both conditions because SV is low.

How Do We Measure Cardiac Output (LO2)?

Now that we understand what CO is, how do we measure it? There are a variety of ways to measure CO.

Clinical Testing

The primary assessment of heart function or cardiac output (CO) is typically noninvasive imaging by **echocardiography**. This procedure can estimate cardiac output effectively and is often the only examination necessary for this purpose.

Under special circumstances, there are a couple of invasive tests that may be used:

- **Angiography** is the measurement of left ventricular (LV) volume during a left-heart catheterization (angiogram) in which dye is injected into the LV. X-rays are then analyzed to calculate the amount of dye remaining in the heart after each beat. This is an invasive test requiring catheter placement in the left heart, so it is used much less often than echocardiography.
- **Thermodilution** is described below. Due to its invasive nature, thermodilution is echocardiography.

Fick's Principle

Fick's principle measures cardiac output (CO) by determining how much oxygen the body consumes in a given time. First, you need to know how much oxygen goes in vs goes out of the tissues. Arterial blood carries oxygen to tissues, while venous blood returns with less oxygen after tissues extract it for their metabolism.

Next, you need to know the oxygen consumption. The loss of oxygen, or consumption by the tissues, also depends on blood flow. By measuring the total oxygen consumption and the difference between arterial and venous oxygen content, we can calculate how much blood must have flowed through the circulatory system.

$$CO = \dot{V}O_2/(C_a - C_v)$$

If we know how much oxygen was consumed in total ($\dot{V}O_2$), and the oxygen content per liter in the arteries vs veins (C_a–C_v), we can calculate how much blood had to flow (CO) through the organs to produce the observed oxygen content difference (see equation above).

So how do we get the values to plug into the equation? Looking at Figure 2, we see that:

- $\dot{V}O_2$ can be direc inspired and expired oxygen concentrations.
- C_a can be measured from the arterial blood (equal to the pulmonary venous blood).
- C_v can be measured by sampling blood from the distal pulmonary artery (equal to the systemic venous blood) by using right-heart catheterization.

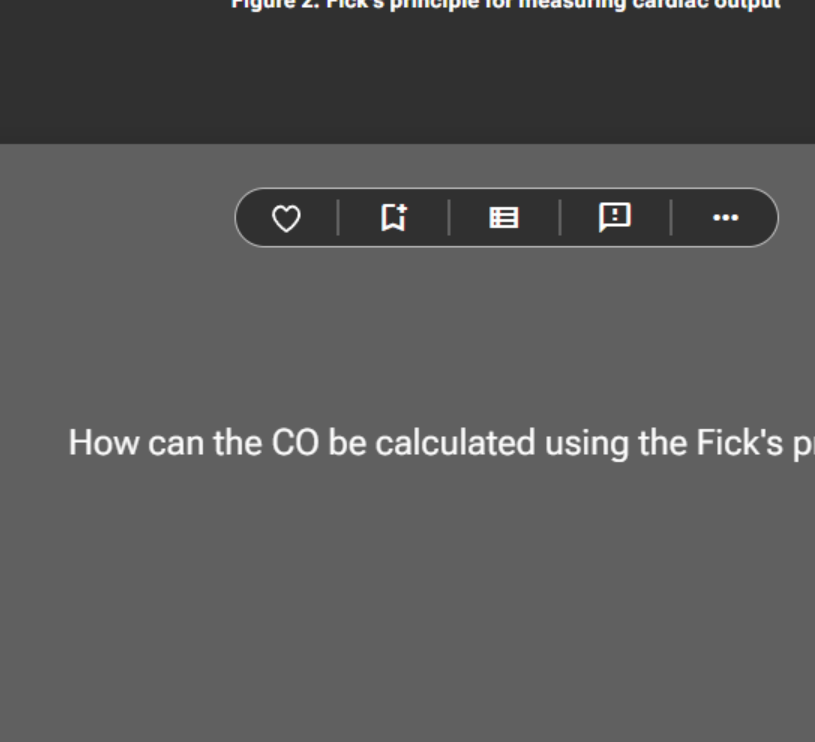
Spirometer
Oxygen consumption (VO₂) = 250mL O₂/min
O₂ uptake
PA
PV
Lung
RV
LV
Aorta
Sample PA oxygen saturation (SaO₂) = 75%
Sample arterial oxygen saturation (SaO₂) = 95%
O₂ release at tissues

Figure 2. Fick's principle for measuring cardiac output

How can the CO be calculated using the Fick's principle?

Dye Dilution Method

The dye dilution method is another way to measure cardiac output (CO). In this technique, a known quantity of dye is injected into the patient's right atrium through a transvenous catheter. The dye concentration is then measured in a peripheral artery over time. The concentration will rise, peak, and fall. A plateau will be reached when the dye circulates back through the heart and returns to the artery. But, if we extrapolate the slope of the initial fall, we can calculate the time that it would take for the bolus of dye to pass through the body (Figure 3).

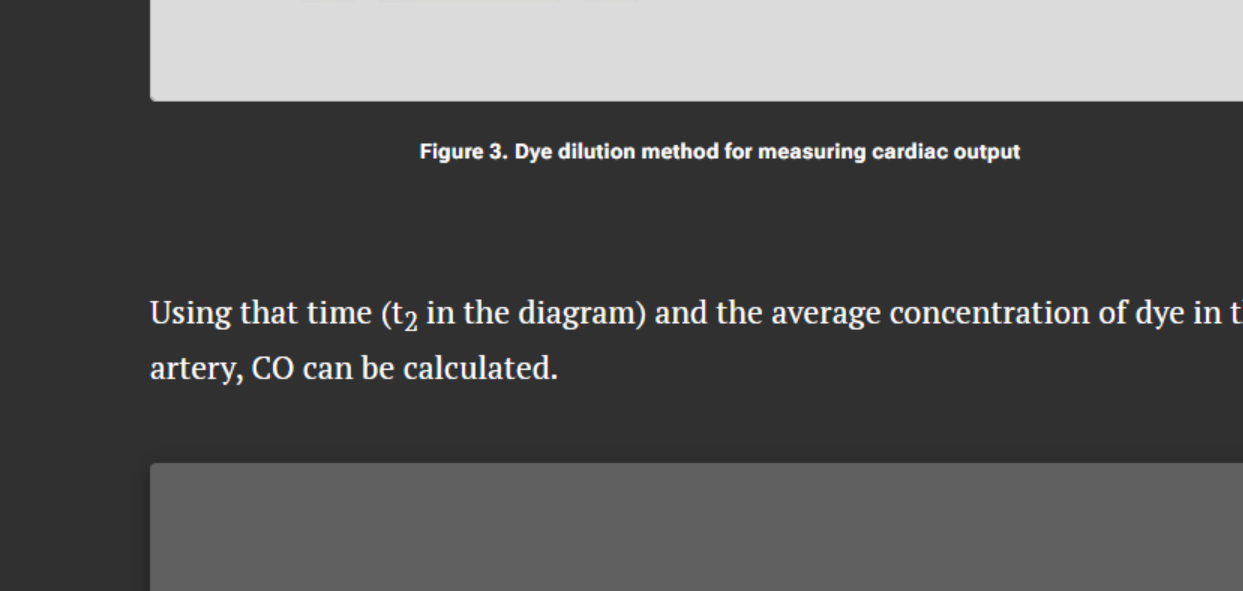

Figure 3. Dye dilution method for measuring cardiac output

Using that time (t_2 in the diagram) and the average concentration of dye in the artery, CO can be calculated.

How is the dye dilution method performed to calculate the CO?

Thermodilution Me

The thermodilution method is used in some intensive care units for the precise calculation of cardiac output (CO). It is similar to dye dilution in that it involves an injection into the right atrium and then measuring arterial blood over time. Instead of dye, though, this method involves injecting a bolus of prechilled blood or cold saline into the heart. We then measure the change in blood temperature over time and calculate the rate of the change of temperature, symbolized by T(t).

How is the thermodilution method used to calculate the CO?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to the patient, TU, how would you respond to her questions? This case highlights the importance of measuring cardiac function (CO and EF) in diagnosing and man

TU's medications are typical for heart failure patients, to reduce the blood volume and force of contraction of the heart. You explain that cardiac output (CO) and ejection fraction (EF) are both measures of heart function, both of which were reduced during her hospitalization, indicating heart failure.

Since she is stable and the symptoms have been resolved with medication, her prognosis is good. You counsel her on the importance of medication adherence and monitoring for signs of worsening heart failure. Given her history, further evaluation—including additional imaging or laboratory tests—may be necessary to determine the underlying cause. You scheduled a follow-up visit in one month.

Summary

- Cardiac output (CO) is the amount of blood that the heart pumps per unit of time; it is the product of stroke volume (SV) and heart rate (HR).
- The stroke volume is the amount of blood ejected per beat and is calculated using end-diastolic volume (EDV) end-systolic volume (ESV): EDV – ESV.
- The ejection fraction (EF) is the percentage of the EDV expelled from the left ventricle (LV) in one heartbeat: EF = SV/ EDV.
- CO can be measured using a variety of methods, including echocardiography (routine in clinical practice), the Fick's principle, dye dilution, and thermodilution.
- The Fick's principle uses oxygen consumption and the difference between the oxygen concentrations in the arteries and veins to calculate the CO.
- Dye dilution involves injecting a dye into the right atrium and then measuring the ch to calculate the CO.
- Thermodilution into the right atrium and then measuring the change in temperature over time to calculate the CO.

Review Questions

1. A female is competing in a triathlon and has her cardiac output (CO) continuously measured non-invasively via estimated continuous CO (eCCO) as she runs. Fifteen minutes in, her CO and stroke volume are measured. Twenty minutes after the first measurement, the same parameters are measured. Her stroke volume has decreased, but her CO has increased. What is the most likely explanation?

- A. ○ End-diastolic volume has decreased
- B. ○ End-systolic volume has increased
- C. ● Heart rate has increased
- D. ○ Preload has decreased
- E. ○ Stroke volume has increased

Hide Explanation

The correct answer is heart rate has increased (C). The two factors that determine cardiac output (CO) are stroke volume and heart rate. If the woman's stroke volume has decreased but he ve increased. End-diastolic volume, end-systolic volume, and preload help determine stroke volume, but they are not direct parts of the formula for CO. End-diastolic volume has decreased (A) and end-systolic volume has increased (B), and preload has decreased (D) would all be associated with decreased stroke volume. We know from the description that her stroke volume has increased (E) is not correct.

2. A male is diagnosed with heart failure with preserved ejection fraction. Through echocardiography, the left ventricular end-diastolic volume is determined to be 90 mL, and the end-systolic volume is determined to be 40 mL. What are the stroke volume and ejection fraction in this patient, respectively?

- A. ○ 40 mL and 44%
- B. ○ 40 mL and 56%
- C. ○ 50 mL and 44%
- D. ● 50 mL and 56%
- E. ○ 90 mL and 44%

Hide Explanation

The correct answer is 50 mL and 55% (D). The stroke volume (SV) can be calculated by subtracting the end-systolic volume (ESV) from the end-diastolic volume (EDV): SV = EDV – ESV = 90 mL – 40 mL = 50 mL. Therefore, 40 mL and 44% (A) and 40 mL and 55% (B) are inco d by dividing the SV by the EDV: EF = SV/EDV = 50 mL/90 mL = 0.55, or 55%, so 50 mL and 44% (C) and 90 mL and 44% (E) are wrong. Notice that the SV is slightly lower than the SV in healthy adults (approximately 70 mL), whereas the EF is within the normal range (55%-60%). This is because in HFpEF, the heart cannot fill properly, resulting in a decrease in EDV. A decrease in EDV leads to a decrease in SV. Because both SV and EDV are decreased, the EF remains unaffected.

3. Which of the following methods of calculating the cardiac output does not require a right-heart catheterization?

- A. ● Angiography
- B. ○ Dye dilution
- C. ○ Fick's principle
- D. ○ Thermodilution

Hide Explanation

The correct answer is **A).** In angiography, a catheter is inserted into a peripheral artery (eg, femoral artery) and subsequently advanced into the left ventricle through the aortic valve, where a radiocontrast dye is injected. The left ventricular end-diastolic and end-systolic volumes are estimated by analyzing x-rays, thus permitting the calculation of the stroke volume and cardiac output. All other methods require catheterization of the right heart, **B and D),** to inject a substance, as in dye dilution and as content per the Fick's principle.

4. A 72-year-old man's cardiac catheterization values are: Pulmonary artery O_2 content = 20 ml O_2/ 100 ml blood; Pulmonary vein O_2 content = 12 ml O_2/100 mL blood, O_2 consumption (VO_2) = 280 ml/min. Stroke Volume = 40 ml. What is the man's cardiac output?

- A. ○ 1.4 L/min
- B. ● 3.5 L/min
- C. ○ 4.67 L.min
- D. ○ 7.0 L/min
- E. ○ 8.0 L/min
- F. ○ 9.24 L/min

Hide Explanation

Correct answer **B).** The cardiac output is calculated using Fick's equation: CO = VO_2/(Ca-Cv). Substituting the given values: CO = 280 /(20 - 12) ; CO = 3500 mL/min = 3.5 L/min. **A).** Incorrectly dividing VO_2 by only the venous O_2 content: 280 × 100 ÷ 12 = 2333 mL/min = 2.33 L/min. **C).** Incorrectly dividing VO_2 by only the arterial O_2 content 280 ÷ 0.2 = 1400 mL/min = 4.67 L/min **D).** Incorrectly dividing VO_2 by stroke volume instead of the L = 7.0 L/min C. **E).** Incorrectly multip on: 40 mL × 280 mL/min ÷ 1000 = 11.2 L/min

5. A patient in the emergency department has an ejection fraction (EF) of 0.4, a heart rate of 95 beats/min, and a cardiac output (CO) of 3.5 L/min. What is the patient's end-diastolic volume (EDV)?

- A. ○ 36.8 mL
- B. ○ 72.3 mL
- C. ● 92.0 mL
- D. ○ 102.0 mL
- E. ○ 145.4 mL

Hide Explanation

Correct answer D). To solve this problem, we need to understand the relationship between cardiac output, stroke volume, heart rate, ejection fraction, and end-diastolic volume. Step 1: Calculate the stroke volume (SV): CO= HR x SV; therefore SV = CO/HR = 3.5/95 = 0.03684 L/beat = 36.8 mL/beat. Step 2: Calculate the end-diastolic volume (EDV): EF = SV/EDV. Therefore, EDV = SV/EF = 36.84/0.4 EDV = 92.0 mL. This is the stroke volume. B)., D)., and E) are all incorrect

How would you rate this Brick?

References

- Huber W, Kraski T, Haller B, et al. Room-temperature vs iced saline indicator injection for transpulmonary thermodilution. *J Crit Care*. 2014;29(6):1133.e7-1133.e14. doi:10.1016/j.jcrc.2014.08.005
- Mikkelsen ME, Gaieski DF, Johnson NJ. Novel tools for hemodynamic monitoring in critically ill patients with shock. UpToDate. Updated September 28, 2020. Accessed September 27, 2022. https://www.uptodate.com/contents/novel-tools-for-hemodynamic-monitoring-in-critically-ill-patients-with-shock

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate between stroke volume and ejection fraction and calculate both.~~

02 ☑ ~~Explain~~ ~~uding the Fick's~~ ~~principl~~ ~~ds.~~

References

- Huber W, Kraski T, Haller B, et al. Room-temperature vs iced saline indicator injection for transpulmonary thermodilution. *J Crit Care*. 2014;29(6):1133.e7-1133.e14. doi:10.1016/j.jcrc.2014.08.005
- Mikkelsen ME, Gaieski DF, Johnson NJ. Novel tools for hemodynamic monitoring in critically ill patients with shock. UpToDate. Updated September 28, 2020. Accessed September 27, 2022. https://www.uptodate.com/contents/novel-tools-for-hemodynamic-monitoring-in-critically-ill-patients-with-shock

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate between stroke volume and ejection fraction and calculate both.~~

02 ☑ ~~Explain methods of measuring cardiac output, including the Fick's principle, dye dilution, and thermodilution methods.~~

Objective Confidence: 2/2